



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-29

Experiment 29. Backward Chaining

Aim

To implement **Backward Chaining** using Prolog.

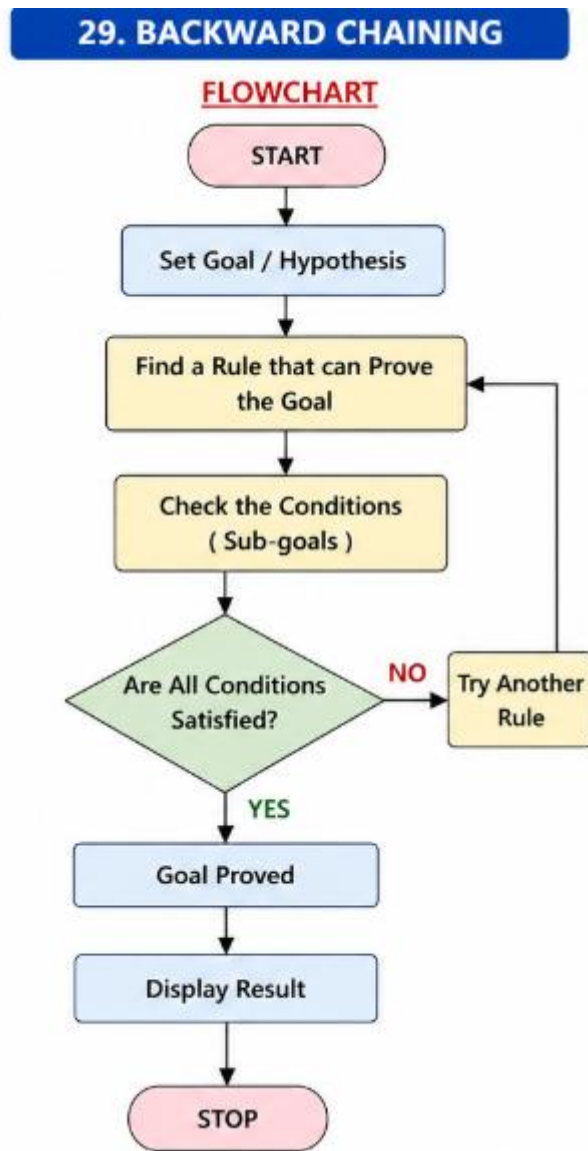
Objectives

- To understand backward reasoning.
- To prove a goal using facts and rules.
- To perform logical inference.

Algorithm

1. Define facts and rules.
2. Set a goal.
3. Find a rule that can prove the goal.
4. Check its conditions.
5. Prove the goal if conditions are satisfied.
6. Display the result.

Flowchart



Code

bird(parrot).

has_wings(parrot).

can_fly(X) :-

bird(X),

```
has_wings(X).
```

```
main :-
```

```
Goal = can_fly(parrot),
```

```
call(Goal),
```

```
write('Goal: '),
```

```
write(Goal), nl,
```

```
write('Result: Goal
```

```
Proved'), nl.
```

```
:- initialization(main).
```

Output

```
bird(parrot).
has_wings(parrot).

can_fly(X) :-
    bird(X),
    has_wings(X).

main :-
    Goal = can_fly(parrot),
    call(Goal),
    write('Goal: '), write(Goal), nl,
    write('Result: Goal Proved'), nl.

:- initialization(main).
```

STDIN

Input for the program (Optic

Output:

Goal: can_fly(parrot)

Result: Goal Proved

Result

Goal: can_fly(parrot)

Result: Goal Proved

Conclusion

Thus, **Backward Chaining** was successfully implemented using Prolog.